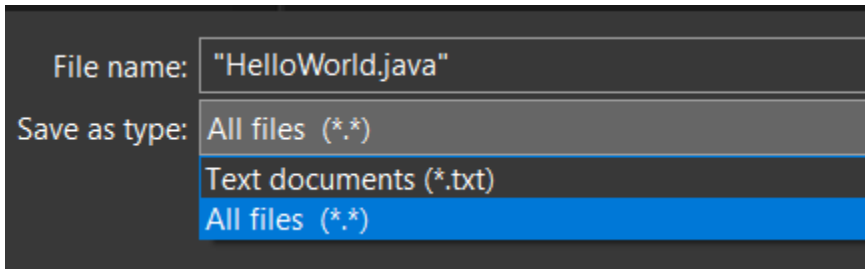


1. Write a simple application like HelloWorld.java. Build and share the image with Docker Hub.

Create a folder DockerJava>> Create a java file as bellows:

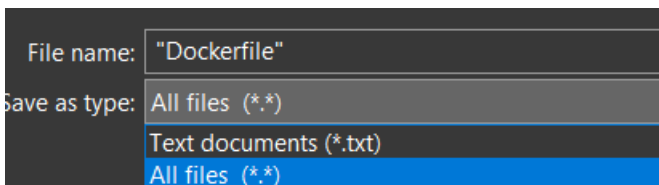


```
public class HelloWorld{
    public static void main(String[] args){
        System.out.println("Hello World!");
    }
}
```

Compile and run on cmd

```
D:\MCA\DevOps\DockerJava>javac HelloWorld.java
D:\MCA\DevOps\DockerJava>java HelloWorld
Hello World!
D:\MCA\DevOps\DockerJava>|
```

Create a “Dockerfile” with no extension as follows:



Code:

Use a lightweight OpenJDK runtime as the base image
FROM eclipse-temurin:17

Set the working directory inside the container
WORKDIR /app

Copy the compiled .class file from your machine to the container
COPY HelloWorld.class .

Execute the application when the container starts
CMD ["java", "HelloWorld"]

Output:

Open cmd from folder location and run:
docker build -t myjavaapp:v1 .

```
D:\MCA\DevOps\DockerJava>docker build -t myjavaapp:v1 .
[+] Building 2.1s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 363B
=> [internal] load metadata for docker.io/library/eclipse-temurin:17
=> [auth] library/eclipse-temurin:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/3] FROM docker.io/library/eclipse-temurin:17@sha256:9cb1a90ce9b7921fe79e544200b0ea1eab76943dd84fb284356ddb
=> [internal] load build context
=> => transferring context: 38B
=> CACHED [2/3] WORKDIR /app
=> CACHED [3/3] COPY HelloWorld.class .
=> exporting to image
=> => exporting layers
=> => writing image sha256:a9dd57a4b8753aebcbb6fd319e876b4ceac9c8926e5be8732e47f1567199dd65
=> => naming to docker.io/library/myjavaapp:v1
View build details: docker:desktop:///dashboard/build/desktop-linux/desktop-linux/mfcy4gqi4rkigygtgihqs5p9x0
```

verifying docker image: docker images

```
D:\MCA\DevOps\DockerJava>docker images
REPOSITORY      TAG          IMAGE ID      CREATED      SIZE
myjavaapp       v1          a9dd57a4b875 21 hours ago 448MB
```

docker run myjavaapp:v1

```
D:\MCA\DevOps\DockerJava>docker run myjavaapp:v1
Hello World!
```

Login to Docker Hub

(Authenticate from browser if asked. Username: kaveri2503, gmail:kaverikadam893)

```
D:\MCA\DevOps\DockerJava>docker login
Authenticating with existing credentials... [Username: kaveri2503]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
```

Login to Docker Hub

```
D:\MCA\DevOps\DockerJava>docker tag myjavaapp:v1 kaveri2503/myjavaapp:v1
```

Push Image to Docker Hub

```
D:\MCA\DevOps\DockerJava>docker push kaveri2503/myjavaapp:v1
The push refers to repository [docker.io/kaveri2503/myjavaapp]
a77025205858: Pushed
4cb4c2c02958: Pushed
3b77731cb571: Mounted from library/eclipse-temurin
31ebd3666df8: Mounted from library/eclipse-temurin
ae3753bea2cc: Mounted from library/eclipse-temurin
7d8c197f9785: Mounted from library/eclipse-temurin
f421a7e99ead: Mounted from library/eclipse-temurin
0c3db79307ab: Mounted from library/eclipse-temurin
v1: digest: sha256:d1cc794275042bf05c6955305666ab08dd5ff30d5dc061463797eee153415540 size: 1990
```

Verify the Image

```
D:\MCA\DevOps\DockerJava>docker run kaveri2503/myjavaapp:v1
Hello World!
```

2. Use docker to modify an existing image, commit and save it as a new one.

Pull existing image

```
D:\MCA\DevOps\DockerJava>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
6f5c5aa4e145: Already exists
1c24335ddd46: Already exists
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
```

Run container >>Create file inside container>>Check file>>Exit container

```
D:\MCA\DevOps\DockerJava>docker run -it ubuntu
root@5acf1fe71148:/# echo "Docker Practical" > test.txt
root@5acf1fe71148:/# cat test.txt
Docker Practical
root@5acf1fe71148:/# exit
exit
```

Check container ID

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
5acf1fe71148	ubuntu	"/bin/bash"	About a minute ago	Exited (0) 6 seconds ago		laughing_banzai
85668742cdfc	kaveri2503/myjavaapp:v1	"/_cacert_entrypoin..."	27 minutes ago	Exited (0) 27 minutes ago		agitated_brahmagupta
e626c53c0477	myjavaapp:v1	"/_cacert_entrypoin..."	51 minutes ago	Exited (0) 51 minutes ago		brave_varahamihira

Commit as New Image (5acf1fe71148 is container id)

```
D:\MCA\DevOps\DockerJava>docker commit 5acf1fe71148 modified-ubuntu:v1
sha256:e5459c1b0285186d363ff62a9eba5806528aeaa1555d3ff634c37082c1fc6273
```

Check images (modified-ubuntu)

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
modified-ubuntu	v1	e5459c1b0285	7 seconds ago	100MB
kaveri2503/myjavaapp	v1	a9dd57a4b875	22 hours ago	448MB
myjavaapp	v1	a9dd57a4b875	22 hours ago	448MB
ubuntu	latest	30ba44506a6d	2 weeks ago	100MB

Run New Image and check file

```
D:\MCA\DevOps\DockerJava>docker run -it modified-ubuntu:v1
root@f074022be39a:/# cat test.txt
Docker Practical
```

3. Write a simple application to create an image, containerize the image, push it to Docker Hub and then remove the images using Docker.

Create a folder DockerJava>> Create a java file as bellows:

File name: "HelloWorld.java"

Save as type: All files (*.*)

Text documents (*.txt)

All files (*.*)

```
public class HelloWorld{
    public static void main(String[] args){
        System.out.println("Hello World!");
    }
}
```

Compile and run on cmd

```
D:\MCA\DevOps\DockerJava>javac HelloWorld.java
```

```
D:\MCA\DevOps\DockerJava>java HelloWorld
Hello World!
```

```
D:\MCA\DevOps\DockerJava>|
```

Create a "Dockerfile" with no extension as follows:

File name: "Dockerfile"

Save as type: All files (*.*)

Text documents (*.txt)

All files (*.*)

Code:

Use a lightweight OpenJDK runtime as the base image
FROM eclipse-temurin:17

Set the working directory inside the container
WORKDIR /app

Copy the compiled .class file from your machine to the container
COPY HelloWorld.class .

Execute the application when the container starts
CMD ["java", "HelloWorld"]

```
D:\MCA\DevOps\DockerJava>docker build -t myapp:v1 .
[+] Building 2.7s (9/9) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 363B 0.0s
=> [internal] load metadata for docker.io/library/eclipse-temurin:17 2.5s
=> [auth] library/eclipse-temurin:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/3] FROM docker.io/library/eclipse-temurin:17@sha256:9cb1a90ce9b7921fe79e544200b0ea1eab76943dd84fb284356ddb 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 38B 0.0s
=> CACHED [2/3] WORKDIR /app 0.0s
=> CACHED [3/3] COPY HelloWorld.class . 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:a9dd57a4b8753aebcbb6fd319e876b4ceac9c8926e5be8732e47f1567199dd65 0.0s
=> => naming to docker.io/library/myapp:v1 0.0s
```

View build details: [docker-desktop://dashboard/build/desktop-linux/desktop-linux/s1emc6qw92beptotgnr21wnrb](https://dashboard.docker.com/build/desktop-linux/desktop-linux/s1emc6qw92beptotgnr21wnrb)

```
D:\MCA\DevOps\DockerJava>docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
modified-ubuntu	v1	e5459c1b0285	About an hour ago	100MB
kaveri2503/myjavaapp	v1	a9dd57a4b875	23 hours ago	448MB
myapp	v1	a9dd57a4b875	23 hours ago	448MB
myjavaapp	v1	a9dd57a4b875	23 hours ago	448MB
ubuntu	latest	30ba44506a6d	2 weeks ago	100MB

```
D:\MCA\DevOps\DockerJava>docker run myapp:v1
Hello World!
```

```
D:\MCA\DevOps\DockerJava>docker login
Authenticating with existing credentials... [Username: kaveri2503]
```

i Info → To login with a different account, run 'docker logout' followed by 'docker login'

```
Login Succeeded
```

```
D:\MCA\DevOps\DockerJava>docker tag myapp:v1 kaveri2503/myapp:v1
```

```
D:\MCA\DevOps\DockerJava>docker push kaveri2503/myapp:v1
The push refers to repository [docker.io/kaveri2503/myapp]
a77025205858: Mounted from kaveri2503/myjavaapp
4cb4c2c02958: Mounted from kaveri2503/myjavaapp
3b77731cb571: Mounted from kaveri2503/myjavaapp
31ebd3666df8: Mounted from kaveri2503/myjavaapp
ae3753bea2cc: Mounted from kaveri2503/myjavaapp
7d8c197f9785: Mounted from kaveri2503/myjavaapp
f421a7e99ead: Mounted from kaveri2503/myjavaapp
0c3db79307ab: Mounted from kaveri2503/myjavaapp
Put "https://registry-1.docker.io/v2/kaveri2503/myapp/manifests/v1": EOF
```

```
D:\MCA\DevOps\DockerJava>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
6885c293bde1	myapp:v1	"/_cacert_entrypoin..."	2 minutes ago	Exited (0) 2 minutes ago		zen_zhukovsky
f074022be39a	modified-ubuntu:v1	"/bin/bash"	About an hour ago	Up About an hour		affectionate_lalande
5acf1fe71148	ubuntu	"/bin/bash"	About an hour ago	Exited (0) About an hour ago		laughing_banzai
85668742cdfc	kaveri2503/myjavaapp:v1	"/_cacert_entrypoin..."	2 hours ago	Exited (0) 2 hours ago		agitated_brahmagupta
e626c53c0477	myjavaapp:v1	"/_cacert_entrypoin..."	2 hours ago	Exited (0) 2 hours ago		brave_varahamihira

Remove container

```
D:\MCA\DevOps\DockerJava>docker rm 6885c293bde1
6885c293bde1
```

Remove local image>> Remove Docker Hub tagged image

```
D:\MCA\DevOps\DockerJava>docker rmi myapp:v1
Untagged: myapp:v1
```

```
D:\MCA\DevOps\DockerJava>docker rmi kaveri2503/myapp:v1
Untagged: kaveri2503/myapp:v1
```

```
D:\MCA\DevOps\DockerJava>docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
modified-ubuntu	v1	e5459c1b0285	About an hour ago	100MB
kaveri2503/myjavaapp	v1	a9dd57a4b875	23 hours ago	448MB
myjavaapp	v1	a9dd57a4b875	23 hours ago	448MB
ubuntu	latest	30ba44506a6d	2 weeks ago	100MB

4. Execute any 5 docker commands and explain their outputs.

1) docker images

```
D:\MCA\DevOps>docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
modified-ubuntu	v1	e5459c1b0285	5 hours ago	100MB
kaveri2503/myjavaapp	v1	a9dd57a4b875	27 hours ago	448MB
myjavaapp	v1	a9dd57a4b875	27 hours ago	448MB
ubuntu	latest	30ba44506a6d	2 weeks ago	100MB

Displays all Docker images available on the local system.

2) docker ps -a

```
D:\MCA\DevOps>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
f074022be39a	modified-ubuntu:v1	"/bin/bash"	5 hours ago	Up 5 hours		affectionate_lalande
5acf1fe71148	ubuntu	"/bin/bash"	5 hours ago	Exited (0) 5 hours ago		laughing_banzai
85668742cdfc	kaveri2503/myjavaapp:v1	"/__cacert_entrypoint..."	5 hours ago	Exited (0) 5 hours ago		agitated_brahmagupta
e626c53c0477	myjavaapp:v1	"/__cacert_entrypoint..."	6 hours ago	Exited (0) 6 hours ago		brave_varahamihira

Displays all containers including stopped containers.

3) docker history myjavaapp:v1

```
D:\MCA\DevOps>docker history myjavaapp:v1
```

IMAGE	CREATED	CREATED BY	SIZE	COMMENT
a9dd57a4b875	27 hours ago	CMD ["java" "HelloWorld"]	0B	buildkit.dockerfile.v0
<missing>	27 hours ago	COPY HelloWorld.class . # buildkit	426B	buildkit.dockerfile.v0
<missing>	27 hours ago	WORKDIR /app	0B	buildkit.dockerfile.v0
<missing>	2 days ago	CMD ["jshell"]	0B	buildkit.dockerfile.v0
<missing>	2 days ago	ENTRYPOINT ["/__cacert_entrypoint.sh"]	0B	buildkit.dockerfile.v0
<missing>	2 days ago	COPY --chmod=755 entrypoint.sh /__cacert_ent...	5.31kB	buildkit.dockerfile.v0
<missing>	2 days ago	RUN /bin/sh -c set -eux; echo "Verifying...	0B	buildkit.dockerfile.v0
<missing>	2 days ago	RUN /bin/sh -c set -eux; ARCH="\$(dpkg --...	281MB	buildkit.dockerfile.v0
<missing>	2 days ago	ENV JAVA_VERSION=jdk-17.0.19+10	0B	buildkit.dockerfile.v0
<missing>	2 days ago	RUN /bin/sh -c set -eux; apt-get update;...	67.1MB	buildkit.dockerfile.v0
<missing>	2 days ago	ENV LANG=en_US.UTF-8 LANGUAGE=en_US:en LC_AL...	0B	buildkit.dockerfile.v0
<missing>	2 days ago	ENV PATH=/opt/java/openjdk/bin:/usr/local/sb...	0B	buildkit.dockerfile.v0
<missing>	2 days ago	ENV JAVA_HOME=/opt/java/openjdk	0B	buildkit.dockerfile.v0
<missing>	2 weeks ago	umoci raw add-layer --image /home/buildd/roc...	179B	Add rock control metadata
<missing>	2 weeks ago	umoci config --image /home/buildd/rockcraft-...	0B	Set annotations
<missing>	2 weeks ago	umoci config --image /home/buildd/rockcraft-...	0B	Set labels
<missing>	2 weeks ago	umoci config --image /home/buildd/rockcraft-...	0B	Set default PATH for bare-based rock
<missing>	2 weeks ago	umoci config --image /home/buildd/rockcraft-...	0B	Set default commands
<missing>	2 weeks ago	umoci config --image /home/buildd/rockcraft-...	0B	Set entrypoint
<missing>	2 weeks ago	umoci raw add-layer --image /home/buildd/roc...	100MB	

Displays history/layers used to build Docker image

4) docker inspect myjavaapp:v1

```
D:\MCA\DevOps>docker inspect myjavaapp:v1
```

```
[
  {
    "Id": "sha256:a9dd57a4b8753aebcbb6fd319e876b4ceac9c8926e5be8732e47f1567199dd65",
    "RepoTags": [
      "kaveri2503/myjavaapp:v1",
      "myjavaapp:v1"
    ],
    "RepoDigests": [
      "kaveri2503/myjavaapp@sha256:d1cc794275042bf05c6955305666ab08dd5ff30d5dc061463797eee153415540"
    ],
    "Parent": "",
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-05-09T09:54:58.232102092Z",
    "DockerVersion": "",
    "Author": "",
    "Config": {
      "Hostname": "",
      "Domainname": "",
      "User": "",
      "AttachStdin": false,
```

Displays detailed information about image, container, network, volume in JSON format.

5) docker pull ubuntu

```
D:\MCA\DevOps>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:f3d28607ddd734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
```

Downloads Docker image from Docker Hub to local system.

5. Deploy a HTML webpage using dockerfile and share it on DockerHub.

Create folder DockerHtml. >> Create "index.html" file (select All files as its extension).

```
<!DOCTYPE html>
<html>
<head>
<title> Docker demo </title>
</head>
<body>
<h1>Hello World </h1>
<p>This is running in a docker container</p>
</body>
</html>
```

Create "Dockerfile"

Code:

```
FROM nginx:stable-alpine3.17-slim
COPY index.html /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

Run

```
D:\MCA\DevOps\DockerHtml>docker build -t mywebapp:v1 .
[+] Building 8.3s (8/8) FINISHED
=> [internal] load build definition from dockerfile
=> => transferring dockerfile: 158B
=> [internal] load metadata for docker.io/library/nginx:stable-alpine3.17-slim
=> [auth] library/nginx:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 204B
=> [1/2] FROM docker.io/library/nginx:stable-alpine3.17-slim@sha256:5893dc08a2cb01e21592ff469346ebaacf49167fbc94
=> => resolve docker.io/library/nginx:stable-alpine3.17-slim@sha256:5893dc08a2cb01e21592ff469346ebaacf49167fbc94
=> => sha256:3c854c8cbf469fda815b8f6183300c07cfa2fbb5703859ca79aff93ae934961b 3.38MB / 3.38MB
=> => sha256:de5d475193dd13b444c2e58fc772d8a3297e370eb90e67e483095bb25f1861a6 1.80MB / 1.80MB
=> => sha256:b407bcc8063852cf7b980fa6d83d6caa2c17b2fa4c10e87835d72f21ed40c41a 625B / 625B
=> => sha256:5893dc08a2cb01e21592ff469346ebaacf49167fbc949f45e1c29111981b0427 9.04kB / 9.04kB
=> => sha256:d0f2d6365b23b7015088c88ee42f119cf585d05558a72d55625e54454c78d3e6 2.11kB / 2.11kB
=> => sha256:ee5112eafd255a2c921521a50e807fef946c83b6d623260ff1d01e856423454 7.15kB / 7.15kB
=> => sha256:da33b1ad0ac4b49641e40469216939f6488c1d8116b2513ba2caa561d4898067 954B / 954B
=> => extracting sha256:3c854c8cbf469fda815b8f6183300c07cfa2fbb5703859ca79aff93ae934961b
=> => sha256:a0fbd691d7c1a07fbdeb8b338578f4e199a49e2491eff171105eb0dc7cf61628 768B / 768B
=> => extracting sha256:de5d475193dd13b444c2e58fc772d8a3297e370eb90e67e483095bb25f1861a6
=> => sha256:16eaaaf5f1c0db05389b2c1c2d90db8c9154289520fad38d580acd2390e846e 1.40kB / 1.40kB
=> => extracting sha256:b407bcc8063852cf7b980fa6d83d6caa2c17b2fa4c10e87835d72f21ed40c41a
=> => extracting sha256:da33b1ad0ac4b49641e40469216939f6488c1d8116b2513ba2caa561d4898067
=> => extracting sha256:a0fbd691d7c1a07fbdeb8b338578f4e199a49e2491eff171105eb0dc7cf61628
=> => extracting sha256:16eaaaf5f1c0db05389b2c1c2d90db8c9154289520fad38d580acd2390e846e
=> [2/2] COPY index.html /usr/share/nginx/html
=> exporting to image
=> => exporting layers
=> => writing image sha256:a7403b4bb9ec22112aa8121141a5de573f438bd5c9183382a43e304d495f4004
=> => naming to docker.io/library/mywebapp:v1
```

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/yfrno18g2x7ogvjdxkmc4uuxi

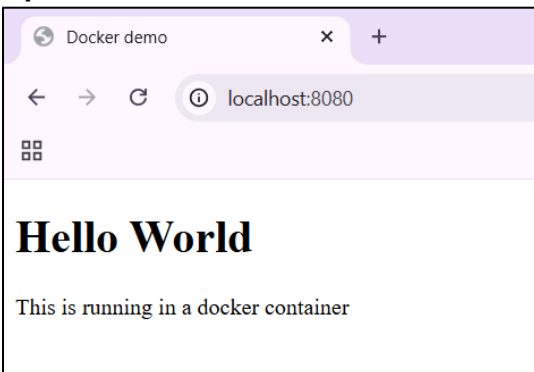

```
D:\MCA\DevOps\DockeHtml>docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
mywebapp	v1	a7403b4bb9ec	5 seconds ago	11.5MB
modified-ubuntu	v1	e5459c1b0285	7 hours ago	100MB
kaveri2503/myjavaapp	v1	a9dd57a4b875	28 hours ago	448MB
myjavaapp	v1	a9dd57a4b875	28 hours ago	448MB
ubuntu	latest	30ba44506a6d	2 weeks ago	100MB

docker run -p 8080:80 mywebapp:v1

```
D:\MCA\DevOps\DockeHtml>docker run -p 8080:80 mywebapp:v1
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/05/10 14:16:20 [notice] 1#1: using the "epoll" event method
2026/05/10 14:16:20 [notice] 1#1: nginx/1.24.0
2026/05/10 14:16:20 [notice] 1#1: built by gcc 12.2.1 20220924 (Alpine 12.2.1_git20220924-r4)
2026/05/10 14:16:20 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/05/10 14:16:20 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/05/10 14:16:20 [notice] 1#1: start worker processes
2026/05/10 14:16:20 [notice] 1#1: start worker process 30
2026/05/10 14:16:20 [notice] 1#1: start worker process 31
2026/05/10 14:16:20 [notice] 1#1: start worker process 32
2026/05/10 14:16:20 [notice] 1#1: start worker process 33
2026/05/10 14:16:20 [notice] 1#1: start worker process 34
2026/05/10 14:16:20 [notice] 1#1: start worker process 35
2026/05/10 14:16:20 [notice] 1#1: start worker process 36
2026/05/10 14:16:20 [notice] 1#1: start worker process 37
2026/05/10 14:16:20 [notice] 1#1: start worker process 38
2026/05/10 14:16:20 [notice] 1#1: start worker process 39
2026/05/10 14:16:20 [notice] 1#1: start worker process 40
2026/05/10 14:16:20 [notice] 1#1: start worker process 41
172.17.0.1 -- [10/May/2026:14:17:32 +0000] "GET / HTTP/1.1" 200 165 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/148.0.0.0 Safari/537.36" "-"
172.17.0.1 -- [10/May/2026:14:17:32 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/148.0.0.0 Safari/537.36" "-"
2026/05/10 14:17:32 [error] 31#31: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referer: "http://localhost:8080/"
```

Open localhost:8080 in browser



```
D:\MCA\DevOps\DockeHtml>docker login
Authenticating with existing credentials... [Username: kaveri2503]
```

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded

Tag image

```
D:\MCA\DevOps\DockeHtml>docker tag mywebapp:v1 kaveri2503/mywebapp:v1
```



```
D:\MCA\DevOps\DockerHtml>docker push kaveri2503/mywebapp:v1
The push refers to repository [docker.io/kaveri2503/mywebapp]
7a2930c50446: Pushed
66afb7c3e6d1: Mounted from library/nginx
439be94dd989: Mounted from library/nginx
08761a690f84: Mounted from library/nginx
419b89bbfa3b: Mounted from library/nginx
5fa2288e0d6e: Mounted from library/nginx
f411324080c: Mounted from library/nginx
v1: digest: sha256:6ab2d215ba711a83c3cdf7b02a37bacac36625417e868e7b715a0baa4a1c7417 size: 1775
```

The screenshot shows the Docker Hub interface for the repository `kaveri2503/mywebapp`. The left sidebar contains navigation links: Repositories, Hardened Images, Collaborations, Settings, Default privacy, Notifications, Billing, Usage, Pulls, and Storage. The main content area shows the repository details, including the name `kaveri2503/mywebapp`, the last pushed time (8 minutes ago), and a list of tags. The 'v1' tag is listed with a pull time of 'less than 1 day' and a push time of '8 minutes'.

6. Write a Dockerfile that explicitly pulls an OpenJDK image before using it as a base, compiles HelloWorld.java, and runs the program inside a containerized environment.

Create a folder as DockerJavaCompile. Open cmd in it.

Create Java Program

```
public class HelloWorld{
    public static void main(String[] args){
        System.out.println("Hello World!");
    }
}
```

Pull OpenJDK Image Explicitly

```
D:\MCA\DevOps\DockerJavaCompile>docker pull eclipse-temurin:17
17: Pulling from library/eclipse-temurin
6f5c5aa4e145: Already exists
1c24335ddd46: Already exists
1074d15b60c9: Already exists
0179dfb07a7d: Already exists
f2355d8d10e0: Already exists
a32c42059121: Already exists
Digest: sha256:9cb1a90ce9b7921fe79e544200b0ea1eab76943dd84fb284356ddb8c96e8f32a
Status: Downloaded newer image for eclipse-temurin:17
docker.io/library/eclipse-temurin:17
```

Create "Dockerfile"

Code:

```
# Pull OpenJDK image as base image
FROM eclipse-temurin:17
```

```
# Set working directory
WORKDIR /app
```

```
# Copy Java source file into container
COPY HelloWorld.java .
```

```
# Compile Java program
RUN javac HelloWorld.java
```

```
# Run Java program
CMD ["java", "HelloWorld"]
```

Build Docker Image

```
D:\MCA\DevOps\DockerJavaCompile>docker build -t javaprogram:v1 .
[+] Building 1.9s (9/9) FINISHED
=> [internal] load build definition from Dockerfile                                docker:desktop-linux 0.1s
=> => transferring dockerfile: 308B                                              0.0s
=> [internal] load metadata for docker.io/library/eclipse-temurin:17            0.0s
=> [internal] load .dockerignore                                                 0.0s
=> => transferring context: 2B                                                  0.0s
=> [1/4] FROM docker.io/library/eclipse-temurin:17                             0.0s
=> [internal] load build context                                                0.1s
=> => transferring context: 166B                                               0.0s
=> CACHED [2/4] WORKDIR /app                                                    0.0s
=> [3/4] COPY HelloWorld.java .                                                 0.0s
=> [4/4] RUN javac HelloWorld.java                                              1.3s
=> exporting to image                                                           0.1s
=> => exporting layers                                                         0.0s
=> => writing image sha256:54d776e2209d468a9d69e8843ed3f764b463d80f20219bcf8368d03754ffd235 0.0s
=> => naming to docker.io/library/javaprogram:v1                             0.0s
```

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/9owv7uod27ven7fxxg2plnlw5

```
D:\MCA\DevOps\DockerJavaCompile>docker run javaprogram:v1
Hello World!
```

```
D:\MCA\DevOps\DockerJavaCompile>docker images
REPOSITORY          TAG          IMAGE ID          CREATED          SIZE
javaprogram         v1          54d776e2209d    About a minute ago  448MB
kaveri2503/mywebapp v1          a7403b4bb9ec    29 minutes ago   11.5MB
mywebapp            v1          a7403b4bb9ec    29 minutes ago   11.5MB
modified-ubuntu     v1          e5459c1b0285    7 hours ago      100MB
kaveri2503/myjavaapp v1          a9dd57a4b875    29 hours ago     448MB
myjavaapp           v1          a9dd57a4b875    29 hours ago     448MB
eclipse-temurin     17          0b4a090763e7    2 days ago       448MB
ubuntu              latest      30ba44506a6d    2 weeks ago      100MB
```

7. Create a Jenkins pipeline to build and deploy a Dockerized Java application (like Exercise-6 mentioned above) automatically. Use GitHub repo to maintain the dockerfile, Jenkinsfile and source code.

Create folder DockerJenkinsApp. >> Create HelloWorld.java

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello from Jenkins Docker Pipeline!");
    }
}
```

Create “DockerFile”

```
FROM eclipse-temurin:17
WORKDIR /app
COPY HelloWorld.java .
RUN javac HelloWorld.java
CMD ["java", "HelloWorld"]
```

Create a github repository as DockerJenkinsApp.

Create "Jenkinsfile".

Code:

```
pipeline {
    agent any

    stages {

        stage('Clone GitHub Repository') {
            steps {
                git 'https://github.com/kaverikadamm/DockerJenkinsApp.git'
            }
        }

        stage('Build Docker Image') {
            steps {
                bat 'docker build -t javajenkinsapp:v1 .'
            }
        }

        stage('Run Docker Container') {
            steps {
                bat 'docker run --name java-container javajenkinsapp:v1'
            }
        }
    }
}
```

Upload in github

```
kaver@KAVERI MINGW64 ~
$ cd d:

kaver@KAVERI MINGW64 /d
$ cd MCA/Devops/DockerJenkinsApp

kaver@KAVERI MINGW64 /d/MCA/Devops/DockerJenkinsApp
$ git init
Initialized empty Git repository in D:/MCA/DevOps/DockerJenkinsApp/.git/

kaver@KAVERI MINGW64 /d/MCA/Devops/DockerJenkinsApp (master)
$ git add .

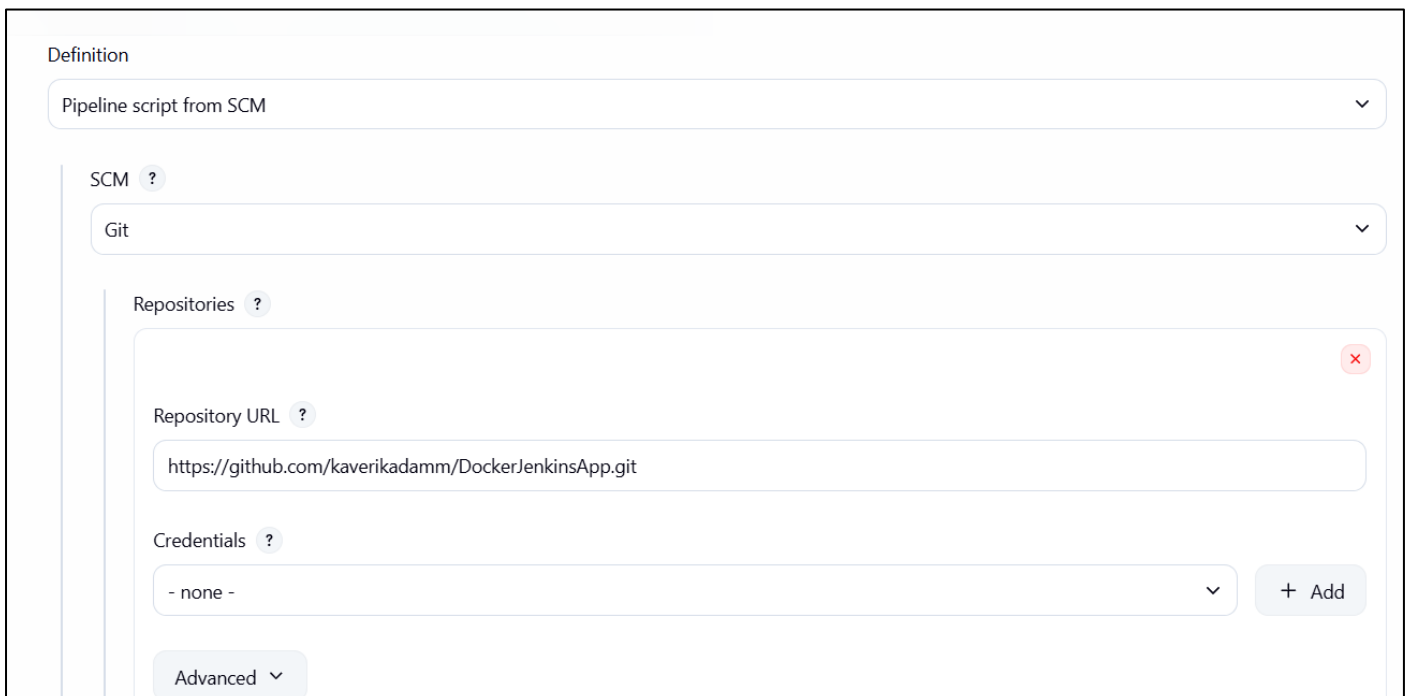
kaver@KAVERI MINGW64 /d/MCA/Devops/DockerJenkinsApp (master)
$ git commit -m "Initial commit - DockerJenkins program"
[master (root-commit) b268ffc] Initial commit - DockerJenkins program
3 files changed, 38 insertions(+)
create mode 100644 Dockerfile
create mode 100644 HelloWorld.java
create mode 100644 jenkinsfile
```

```
kaver@KAVERI MINGW64 /d/MCA/Devops/DockerJenkinsApp (master)
$ git remote add origin https://github.com/kaverikadamm/DockerJenkinsApp.git

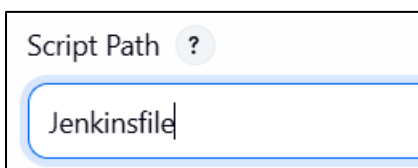
kaver@KAVERI MINGW64 /d/MCA/Devops/DockerJenkinsApp (master)
$ git push -u origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 741 bytes | 741.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kaverikadamm/DockerJenkinsApp.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

Open jenkins(localhost:8090)

Create jenkins pipeline DockerJavaPipeline >> Pipeline script from SCM.>> Enter Repository URL.



The screenshot shows the Jenkins Pipeline Configuration page. The 'Definition' dropdown is set to 'Pipeline script from SCM'. The 'SCM' dropdown is set to 'Git'. Under 'Repositories', there is a single repository entry with the URL 'https://github.com/kaverikadamm/DockerJenkinsApp.git'. The 'Credentials' dropdown is set to '- none -'. There is an '+ Add' button next to the credentials dropdown. An 'Advanced' dropdown is visible at the bottom of the repository list.



The screenshot shows the 'Script Path' field in the Jenkins Pipeline Configuration page. The field is labeled 'Script Path' and contains the text 'Jenkinsfile'.

Output:

```
Started by user Devops
Obtained Jenkinsfile from git https://github.com/kaverikadamm/DockerJenkinsApp.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\DockerJavaPipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/kaverikadamm/DockerJenkinsApp.git
  > C:\Program Files\Git\bin\git.exe init C:\ProgramData\Jenkins\.jenkins\workspace\DockerJavaPipeline # timeout=10
Fetching upstream changes from https://github.com/kaverikadamm/DockerJenkinsApp.git
  > C:\Program Files\Git\bin\git.exe --version # timeout=10
  > git --version # 'git version 2.53.0.windows.1'
  > C:\Program Files\Git\bin\git.exe fetch --tags --force --progress -- https://github.com/kaverikadamm/DockerJenkinsApp.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
  > C:\Program Files\Git\bin\git.exe config remote.origin.url https://github.com/kaverikadamm/DockerJenkinsApp.git # timeout=10
  > C:\Program Files\Git\bin\git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
```

```
  > C:\Program Files\Git\bin\git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision b268ffcba077557e4ca22b054169789ecb02cf60 (refs/remotes/origin/master)
  > C:\Program Files\Git\bin\git.exe config core.sparsecheckout # timeout=10
  > C:\Program Files\Git\bin\git.exe checkout -f b268ffcba077557e4ca22b054169789ecb02cf60 # timeout=10
Commit message: "Initial commit - DockerJenkins program"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone GitHub Repository)
[Pipeline] git
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
  > C:\Program Files\Git\bin\git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\DockerJavaPipeline\.git # timeout=10
Fetching changes from the remote Git repository
  > C:\Program Files\Git\bin\git.exe config remote.origin.url https://github.com/kaverikadamm/DockerJenkinsApp.git # timeout=10
Fetching upstream changes from https://github.com/kaverikadamm/DockerJenkinsApp.git
  > C:\Program Files\Git\bin\git.exe --version # timeout=10
  > git --version # 'git version 2.53.0.windows.1'
  > C:\Program Files\Git\bin\git.exe fetch --tags --force --progress -- https://github.com/kaverikadamm/DockerJenkinsApp.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
```

```
  > C:\Program Files\Git\bin\git.exe rev-parse "refs/remotes/origin/master^{commit}" # timeout=10
Checking out Revision b268ffcba077557e4ca22b054169789ecb02cf60 (refs/remotes/origin/master)
  > C:\Program Files\Git\bin\git.exe config core.sparsecheckout # timeout=10
  > C:\Program Files\Git\bin\git.exe checkout -f b268ffcba077557e4ca22b054169789ecb02cf60 # timeout=10
  > C:\Program Files\Git\bin\git.exe branch -a -v --no-abbrev # timeout=10
  > C:\Program Files\Git\bin\git.exe checkout -b master b268ffcba077557e4ca22b054169789ecb02cf60 # timeout=10
Commit message: "Initial commit - DockerJenkins program"
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build Docker Image)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\DockerJavaPipeline>docker build -t javajenkinsapp:v1 .
#0 building with "default" instance using docker driver

#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 161B 0.0s done
#1 DONE 0.1s

#2 [internal] load metadata for docker.io/library/eclipse-temurin:17
#2 DONE 0.0s
```

```
#3 [internal] load .dockerignore
#3 transferring context: 2B done
#3 DONE 0.0s

#4 [1/4] FROM docker.io/library/eclipse-temurin:17
#4 DONE 0.0s

#5 [internal] load build context
#5 transferring context: 193B 0.0s done
#5 DONE 0.1s

#6 [2/4] WORKDIR /app
#6 CACHED

#7 [3/4] COPY HelloWorld.java .
#7 DONE 0.0s

#8 [4/4] RUN javac HelloWorld.java
#8 DONE 0.8s

#9 exporting to image
#9 exporting layers 0.0s done
#9 writing image sha256:93ccb30da03b6b6a8cabb1081024f5e4d4332a51f39fb0c75fde84da08d6730d done
#9 naming to docker.io/library/javajenkinsapp:v1 done
#9 DONE 0.1s
[Pipeline] }
[Pipeline] // stage
```

```
C:\ProgramData\Jenkins\.jenkins\workspace\DockerJavaPipeline>docker run --name java-container javajenkinsapp:v1
Hello from Jenkins Docker Pipeline!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```